

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech / Ph.D Assessment-I (26th Sept. 2022)

Course (Subject) : **Advanced Optimization Techniques** Course Code : **AM-603 /**
Computer Oriented Optimization Techniques **AM-603D**

Duration : 01 hr

Max Marks : 10

Note. Attempt all questions from the following

1. a). Write the Phases of O.R. Model.
b). Write the Simplex method algorithm.

2. Use Big-M method to solve the following L.P.P

Minimize $Z = 2x_1 + x_2$

S.to. $3x_1 + x_2 = 3$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

3. Use Two-Phase method to solve the following L.P.P

Minimize $Z = x_1 - 2x_2 - 3x_3$

S.to. $-2x_1 + x_2 + 3x_3 = 2$

$$2x_1 + 3x_2 + 4x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$